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Computer Methods in Combustion

Ignition Delay, Peak Combustion Parameters and Emissions of Hydrogen-Enriched Methane (HCNG) Blends in a Constant-Volume Combustion Chamber

A Cantera-based study in the context of a spark-ignition piston engine

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Abstract

This report investigates the combustion behaviour of hydrogen-enriched methane (HCNG) blends in a constant-volume, adiabatic combustion chamber, treated as a zero-dimensional model of autoignition in a spark-ignition piston engine. Using the open-source chemical kinetics library Cantera and the GRI-Mech 3.0 reaction mechanism, the autoignition delay time, the peak combustion temperature and pressure, an idealized Otto-cycle estimate of power and efficiency, and the equilibrium pollutant emissions (NO_x , CO, CO_2) were computed for fuels containing 0–30 % hydrogen by mole. Pure methane (0 % H_2) was used as the conventional-fuel baseline. The results reproduce the well-known trade-off of hydrogen enrichment: shorter ignition delay, higher peak temperature and pressure, a small gain in indicated efficiency, a reduction of CO_2 emissions of about 7 % at 30 % H_2 , and an increase of thermal NO_x of a comparable magnitude.

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1 Introduction

The decarbonisation of transport has renewed interest in gaseous fuels and, in particular, in adding hydrogen to conventional hydrocarbon fuels. Blending hydrogen with methane (natural gas) produces a fuel commonly referred to as HCNG (hydrogen-enriched compressed natural gas), which can be burned in existing spark-ignition (SI) piston engines with only minor modifications. Hydrogen has a high laminar flame speed, a wide flammability range and contains no carbon, so even a moderate hydrogen fraction can improve combustion stability and reduce carbon-based emissions.

The aim of this project is to quantify, by means of chemical-kinetics simulation, how the hydrogen fraction in the fuel affects:

- the autoignition delay time of the mixture,
- the peak combustion temperature and pressure,
- the indicated power and thermal efficiency of an idealized engine cycle,
- the emissions of the main pollutants (NO_x , CO, CO_2).

All quantities are compared against a pure-methane baseline, which represents the conventional fuel. The calculations are performed with Cantera, an open-source suite for thermodynamics, chemical kinetics and transport processes.

2 Literature Review

Hydrogen enrichment of hydrocarbon fuels has been studied extensively in the combustion literature. The reported effects are consistent across many works: adding hydrogen shortens the ignition delay and increases the laminar flame speed, because hydrogen oxidation rapidly builds up the radical pool (H, O, OH) that drives the chain-branching chemistry of the hydrocarbon. The higher reactivity improves lean-burn capability and reduces cycle-to-cycle variability in SI engines.

The carbon content of the fuel decreases monotonically with hydrogen addition, which directly lowers CO_2 and, usually, CO and unburned hydrocarbon emissions. The main drawback is an increase in nitrogen oxides: hydrogen raises the peak flame temperature, and because thermal NO_x formation through the extended Zeldovich mechanism is exponentially sensitive to temperature, even a small temperature rise can noticeably increase NO_x . This CO_2 -versus- NO_x trade-off is the central practical issue of HCNG operation and motivates the parametric study presented here.

For the chemistry, GRI-Mech 3.0 is a widely validated mechanism developed for natural-gas combustion. It contains 53 species and 325 reactions and includes detailed H_2/O_2 chemistry as well as the nitrogen sub-mechanism required to predict NO_x , which makes it well suited to methane-hydrogen mixtures.

3 Model Description

3.1 Combustion chamber and governing assumptions

The fuel-air mixture is modelled in a constant-volume, adiabatic, zero-dimensional reactor. This is the standard idealisation of the autoignition of the unburned charge in a piston engine and is analogous to a constant-volume combustion bomb or a rapid-compression-machine test cell. The reactor is homogeneous (perfectly stirred), there is no heat loss to the walls and no work exchange, so the internal energy and the volume are constant during the reaction. The parameters of the research chamber used throughout this work are summarised in Table 1.

Table 1: Definition of the research combustion chamber and operating point.

Quantity	Value
Chamber type	constant-volume, adiabatic, 0-D reactor
Chamber volume	0.5 L ($5 \times 10^{-4} \text{ m}^3$)
Initial temperature T_0	1000 K
Initial pressure p_0	20 atm ($\approx 20.3 \text{ bar}$)
Equivalence ratio ϕ	1.0 (reference)
Oxidizer	air, $\text{O}_2:\text{N}_2 = 1 : 3.76$
Fuel	$\text{CH}_4 + \text{H}_2$, $x_{\text{H}_2} = 0, 0.1, 0.2, 0.3$
Reaction mechanism	GRI-Mech 3.0 (53 species)

3.2 Mixture definition

The fuel is a mixture of methane and hydrogen, defined by the hydrogen mole fraction x_{H_2} in the fuel:

$$\text{fuel} = x_{\text{H}_2} \text{H}_2 + (1 - x_{\text{H}_2}) \text{CH}_4. \quad (1)$$

The fuel and air are mixed at a prescribed equivalence ratio ϕ , defined as

$$\phi = \frac{(F/A)}{(F/A)_{\text{stoich}}}, \quad (2)$$

where F/A is the fuel-to-air ratio. Values $\phi < 1$ correspond to lean mixtures and $\phi > 1$ to rich ones.

3.3 Ignition delay

The autoignition delay τ is obtained by integrating the governing equations of the constant-volume reactor in time and detecting the instant of the steepest temperature rise:

$$\tau = \arg \max_t \frac{dT}{dt}. \quad (3)$$

This maximum-gradient criterion is a standard and robust definition of the ignition delay. The temperature dependence of τ follows an Arrhenius-type law and is therefore presented on a $\log \tau$ versus $1000/T$ plot.

3.4 Peak temperature and pressure

The maximum (adiabatic) state reached by the burned gas is obtained by equilibrating the mixture at constant internal energy and volume (a UV equilibrium), which yields the peak temperature T_{max} and peak pressure P_{max} without time integration.

3.5 Power and efficiency estimate

An approximate, idealized Otto cycle is used to translate the chemistry into engine-level quantities. The chemical energy available per charge is computed as the constant-volume heating value of the mixture, i.e. the drop in internal energy when the products are returned to the initial temperature at constant volume:

$$Q_{\text{in}} = m (u_{\text{reactants}}(T_0) - u_{\text{products}}(T_0)), \quad (4)$$

where m is the charge mass. The ideal Otto-cycle thermal efficiency depends on the compression ratio r and the ratio of specific heats γ of the fresh charge:

$$\eta = 1 - \left(\frac{1}{r}\right)^{\gamma-1}. \quad (5)$$

The indicated work per cycle is $W = Q_{\text{in}} \eta$ and the indicated power follows from the engine speed (here $r = 10$, 3000 rpm, four-stroke). This estimate is intended for relative comparison between fuels; it neglects heat losses, friction and finite burn duration.

3.6 Emissions

The pollutant content of the burned gas is evaluated from the equilibrium composition of the products at the peak (UV) state. Concentrations are reported both as exhaust mole fractions (ppm, vol.%) and as emission indices, i.e. the mass of pollutant per kilogram of fuel:

$$\text{EI}_i = \frac{Y_i}{Y_{\text{fuel}}} \times 1000 \text{ g kg}^{-1}. \quad (6)$$

It should be noted that these are equilibrium values at the peak temperature; they represent an upper bound and are best interpreted as a relative comparison between the fuels rather than as tailpipe concentrations.

4 Results

4.1 Ignition delay

Figure 1 shows the ignition delay as a function of the initial temperature. The expected Arrhenius behaviour is recovered: the delay falls by several orders of magnitude as the temperature increases. Increasing the hydrogen fraction shifts the curves downward, i.e. hydrogen consistently shortens the ignition delay, with the strongest relative effect at lower temperatures.

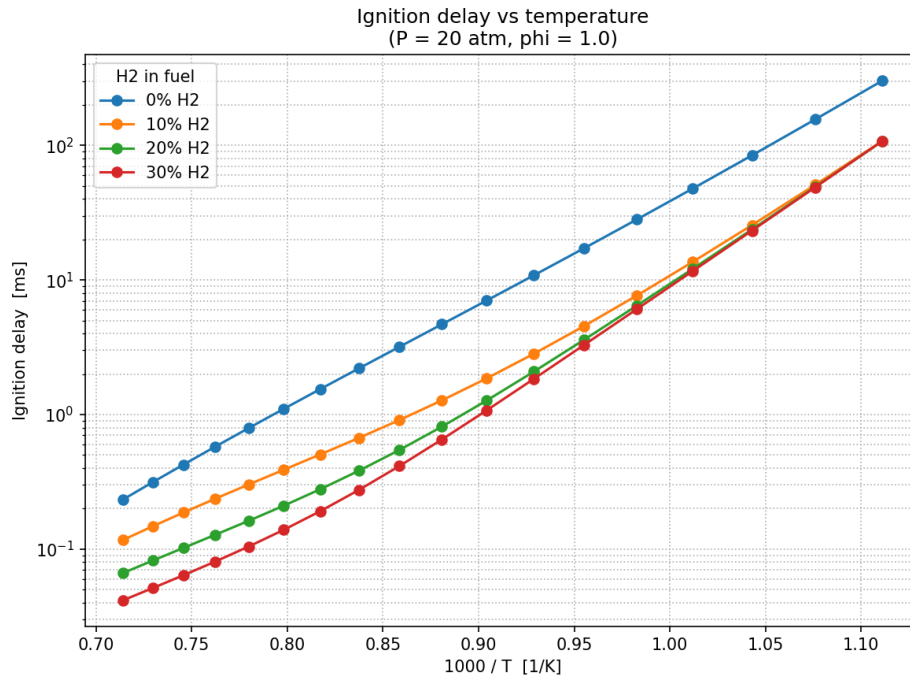


Figure 1: Ignition delay versus initial temperature (Arrhenius plot) for different hydrogen fractions at $p_0 = 20 \text{ atm}$, $\phi = 1$.

The dependence on initial pressure and on equivalence ratio is shown in Figures 2a and 2b. A higher pressure shortens the delay, and the shortest delays occur for slightly rich mixtures; in both cases hydrogen enrichment lowers the delay further.

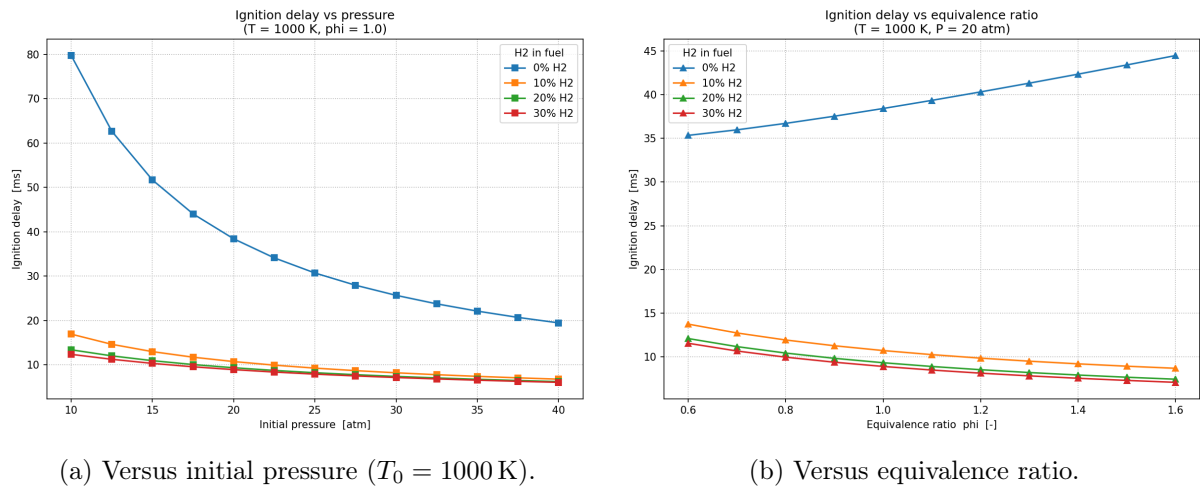


Figure 2: Ignition delay versus pressure and equivalence ratio for different hydrogen fractions.

4.2 Peak temperature and pressure

The peak combustion temperature and pressure as functions of the equivalence ratio are shown in Figures 3a and 3b. Both quantities peak slightly on the rich side ($\phi \approx 1.05$ – 1.1) and increase with hydrogen content. The higher peak temperature is the direct cause of the increased NO_x discussed below.

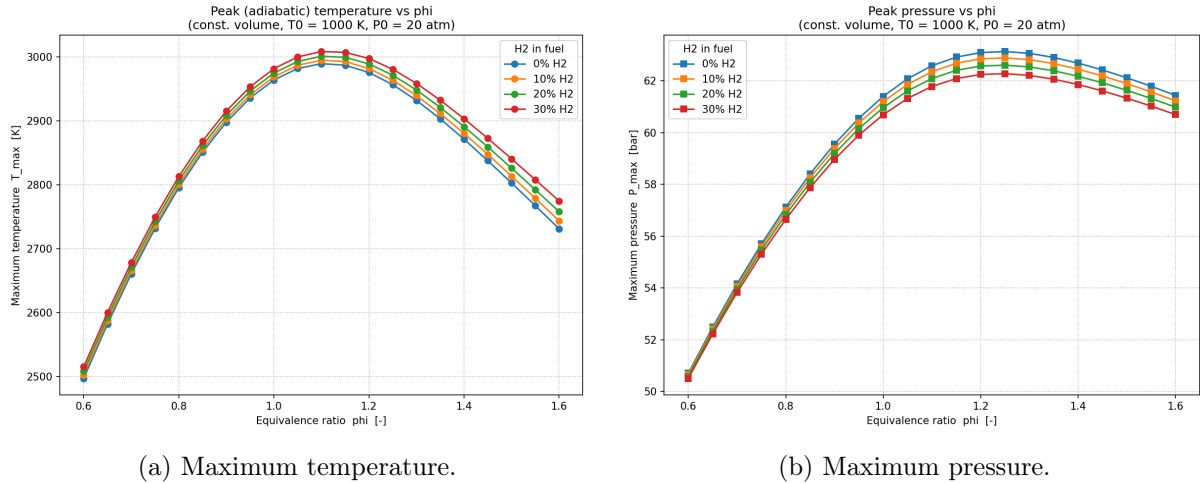


Figure 3: Peak (adiabatic, constant-volume) temperature and pressure versus equivalence ratio for different hydrogen fractions.

4.3 Power and efficiency

Figure 4 presents the change in indicated power and thermal efficiency relative to pure methane, obtained from the idealized Otto cycle of Eq. (5). Hydrogen enrichment raises the ratio of specific heats of the charge, which slightly improves the cycle efficiency; at 30 % H_2 the estimated efficiency gain is about +1.4 % and the power gain about +0.8 %. These are modest because, at a fixed operating point and chamber volume, it is mainly the mixture composition (not the total energy content) that changes.

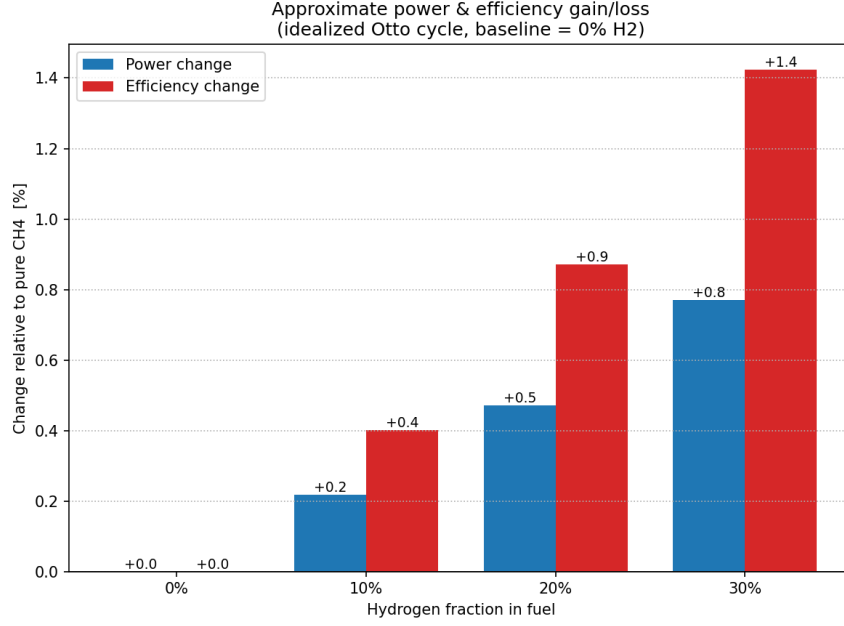


Figure 4: Approximate change in indicated power and thermal efficiency relative to pure methane (idealized Otto cycle, $r = 10$).

4.4 Emissions

The equilibrium emissions at the reference operating point are listed in Table 2. The corresponding percentage changes relative to the conventional fuel are plotted in Figure 5a, and the equivalence- ratio dependence of NO_x is shown in Figure 5b.

Table 2: Exhaust emissions at the reference point ($T_0 = 1000 \text{ K}$, $p_0 = 20 \text{ atm}$, $\phi = 1$). EI = emission index (g per kg of fuel).

x_{H_2} [%]	NO_x [ppm]	CO [ppm]	CO_2 [vol%]	EI_{NO_x} [g/kg]	EI_{CO} [g/kg]	EI_{CO_2} [g/kg]
0	9250	25543	6.74	186.2	479.6	1989.7
10	9274	25189	6.50	189.7	480.7	1950.2
20	9300	24752	6.23	193.9	481.6	1903.0
30	9329	24200	5.90	199.1	482.0	1845.7
Change at 30 % H_2 vs. 0 %:				+7.0 %	+0.5 %	-7.2 %

The carbon-based emissions decrease with hydrogen content: CO_2 drops by about 7% at 30% H_2 , simply because the fuel contains less carbon. At the same time, NO_x rises by a similar amount, driven by the higher peak temperature. CO is almost unchanged at this operating point.

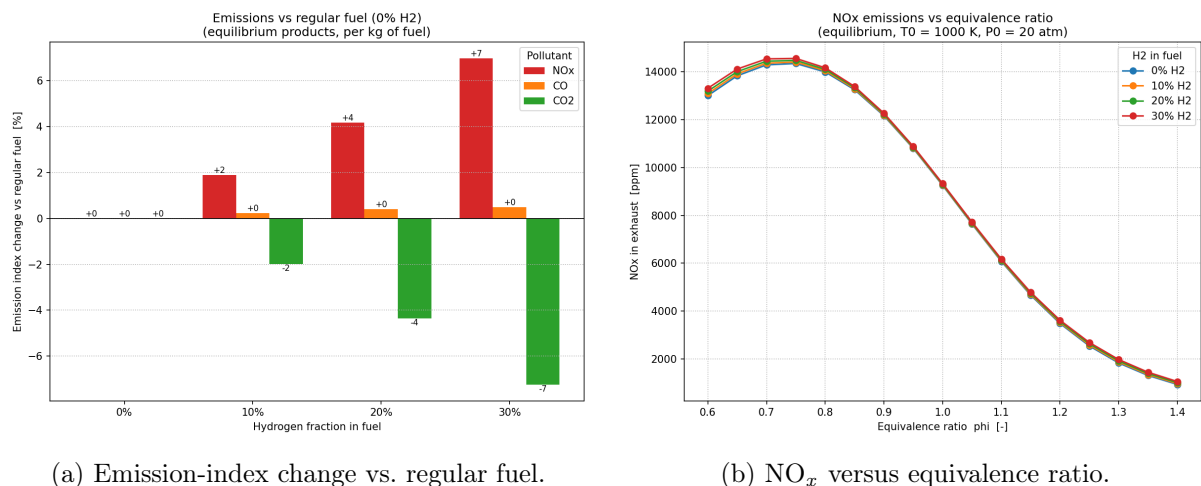


Figure 5: Emissions of HCNG blends: percentage change relative to the conventional fuel (left) and the equivalence-ratio dependence of NO_x (right). NO_x peaks for slightly lean mixtures, as expected from thermal NO_x chemistry.

5 Summary and Conclusions

A zero-dimensional, constant-volume Cantera model was used to study the combustion of hydrogen-enriched methane (HCNG) blends in the context of a spark-ignition piston engine. The main findings are:

- Hydrogen enrichment consistently shortens the autoignition delay; the effect is strongest at low temperature and high pressure. In an engine this implies improved ignitability but also an increased knock tendency that would have to be managed.
- The peak combustion temperature and pressure increase with hydrogen content and are highest for slightly rich mixtures.
- The idealized Otto-cycle estimate predicts a small efficiency gain ($\approx +1.4\%$) and power gain ($\approx +0.8\%$) at 30 % H_2 , due to the higher ratio of specific heats of the charge.
- Hydrogen enrichment lowers CO_2 emissions by about 7 % at 30 % H_2 , but increases thermal NO_x by a comparable amount. This CO_2 -versus- NO_x trade-off is the key practical result.

Overall, moderate hydrogen addition is an effective route to lower carbon-based emissions in conventional SI engines, at the cost of higher NO_x that would require lean operation or after-treatment to control. The quantitative emission values reported here are equilibrium estimates and should be treated as relative trends rather than absolute tailpipe figures.

References

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